The propagation mode (2) contour is a circle with a radius equal to the propagation mode (2) required distance. Unlike the case for propagation mode (1), the propagation mode (2) contour is not centred on the earth station's physical location, instead it is centred on a point on the earth's surface immediately below the centre of the common volume.

A common volume can exist, with equal probability, at any point along the earth station beam between the earth station's location and the point at which the beam reaches the rain height. To provide appropriate protection for/from terrestrial stations[8], the centre of the common volume is assumed to be half way between the earth station and the point at which its beam intersects the rain height. The distance between the projection of this point on to the Earth's surface and the location of the earth station is known as Δd (see § 4 of Annex 2). The centre of the propagation mode (2) contour is therefore Δd (km) from the earth station on the azimuth of the earth station's main beam axis.

5.1 The required distance for propagation mode (2)

Propagation mode (2) required distances are measured along a radial originating at the centre of the rain scatter common volume. The calculation requires iteration for distance, starting at the same minimum distance as that defined for propagation mode (1) until either the required propagation mode (2) minimum required loss, or a latitude-dependent propagation mode (2) maximum calculation distance, is achieved. The propagation mode (2) calculations use the method described in Annex 2. The calculations only need to be performed in the frequency range 1 000 MHz to 40.5 GHz. Outside this frequency range, rain scatter interference can be neglected and the propagation mode (2) required distance is set to the minimum coordination distance given by equations (13) to (18).

ANNEX 1

Determination of the required distance for propagation mode (1)

1 Adjustments for earth station horizon elevation angle and distance

For propagation mode (1), the required distance depends on the characteristics of the physical horizon around the earth station. The horizon is characterized by the horizon distance d_h (see below), and the horizon elevation angle ε_h. The horizon elevation angle is defined here as the angle (degrees), viewed from the centre of the earth station antenna, between the horizontal plane and a ray that grazes the physical horizon in the direction concerned. The value of ε_h is positive when the physical horizon is above the horizontal plane and negative when it is below.

It is necessary to determine horizon elevation angles and distances for all azimuths around an earth station. In practice it will generally suffice to do this in azimuth increments of 5°. However, every attempt should be made to identify, and take into consideration, minimum horizon elevation angles that may occur between those azimuths examined in 5° increments.

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[8] This procedure does not apply for the case of an earth station sharing a frequency band with other earth stations operating in the opposite direction of transmission, as for that specific case the propagation mode (2) contour is based on a geometric construction.

For the purposes of the determination of the propagation mode (1) required distance it is useful to separate the propagation effects related to the local horizon around the earth station which, on some or all azimuths, may be determined by nearby hills or mountains, from the propagation effects on the remainder of the path. This is achieved by referencing the propagation model to a 0° horizon elevation angle for the coordinating earth station, and then to include a specific term A_h to deal with the known horizon characteristics of the earth station being coordinated. Where appropriate, A_h modifies the value of the path loss, on each azimuth, from which the propagation mode (1) required distance is derived.

There are two situations in which the level of attenuation for the propagation mode (1) path loss with respect to the reference 0° case can change:

– The first is where the coordinating earth station has a positive horizon elevation angle (on a particular azimuth). In this case, it will benefit from additional diffraction propagation losses over the horizon (generally referred to as site shielding). As a result, the attenuation A_h is positive and the value of the required path loss is reduced, with respect to the reference 0° horizon elevation angle case (see equations (27a) and (27b)).

– The second situation is where the coordinating earth station is at a location above the local foreground, and has a negative (downward) horizon elevation angle on a particular azimuth. In this case, a measure of additional protection is necessary because the path angular distance along the radial is reduced and hence the path loss for a given distance will be lower than for the zero degree elevation angle case. It is convenient to deal with this effect as part of the site shielding calculation. As a result, the attenuation A_h will be negative and the value of the required path loss is increased, with respect to the reference 0° horizon elevation angle case.

The contribution made by the attenuation arising from the coordinating earth station's horizon characteristics to the propagation mode (1) minimum required loss modifies the value of path loss that then needs to be determined in the three propagation mode (1) models. The attenuation A_h is calculated for each azimuth around the coordinating earth station as follows.

The distance of the horizon, d_h, from the earth station's location, is determined by:

$$d_h = \begin{cases} 0.5 \text{ km} & \text{if no information is available about the horizon distance, or if the distance is} < 0.5 \text{ km} \\ \text{horizon distance (km)} & \text{if this is within the range } 0.5 \text{ km} \le \text{horizon distance} \le 5.0 \text{ km} \\ 5.0 \text{ km} & \text{if the horizon distance is} > 5.0 \text{ km} \end{cases}$$

The contribution made by the horizon distance, d_h, to the total site shielding attenuation is given by A_d (dB) for each azimuth using:

$$A_d = 15\left[1 - \exp\left(\frac{0.5 - d_h}{5}\right)\right]\left[1 - \exp\left(-\varepsilon_h f^{1/3}\right)\right] \quad \text{dB} \qquad (23)$$

where f is the frequency (GHz) throughout this Annex.

The total site shielding attenuation along each azimuth from the coordinating earth station is given by:

$$A_h = \begin{cases} 20 \log (1 + 4.5\varepsilon_h f^{1/2}) + \varepsilon_h f^{1/3} + A_d & \text{dB for } \varepsilon_h \geq 0^\circ \quad (24a) \\ 3\left[(f+1)^{1/2} - 0.0001\, f - 1.0487\right] \varepsilon_h & \text{dB for } 0^\circ > \varepsilon_h \geq -0.5^\circ \quad (24b) \\ -1.5\left[(f+1)^{1/2} - 0.0001\, f - 1.0487\right] & \text{dB for } \varepsilon_h < -0.5^\circ \quad (24c) \end{cases}$$

The value of A_h must be limited to satisfy the conditions:

$$-10 \leq A_h \leq (30 + \varepsilon_h) \quad (25)$$

In equations (23), (24) and (25) the value of ε_h must always be expressed in degrees. The limits defined in equation (25) are specified because protection outside these limits may not be realized in practical situations.

2 Frequencies between 100 MHz and 790 MHz

The propagation model given in this section is limited to an average annual time percentage, p, in the range 1% to 50%.

An iterative process is used to determine the propagation mode (1) required distance. First, equation (27) is evaluated. Then, commencing at the minimum coordination distance, d_{min}, given by the method described in § 1.5.3 of the main body of this Appendix, equations (28) to (31) are iterated for distances d_i (where i = 0, 1, 2,...) incremented in steps of s (km) as described in § 1.3 of the main body of this Appendix. In each iteration, d_i is the distance considered. This process is continued until either of the following expressions becomes true:

$$L_2(p) \geq \begin{cases} L_1(p) & \text{for the main or supplementary contour} \\ L_{1q}(p) & \text{for the auxiliary contour} \end{cases} \quad (26a)$$

or:

$$d_i \geq \begin{cases} d_{max1} & \text{for the main or supplementary contour} \\ d_1 & \text{for the auxiliary contour} \end{cases} \quad (26b)$$

The required distance, d_1, or the auxiliary contour distance, d_q, are then given by the distance for the last iteration: i.e.

$$d_1 = d_i \quad (26c)$$

or:

$$d_q = d_i \quad (26d)$$

As the eventual mix of zones along a path is unknown, all paths are treated as if they are potential land and sea paths. Parallel calculations are undertaken, the first assuming the path is all land and a second assuming it is all sea. A non-linear interpolation is then performed, the output of which depends upon the current mix of land and sea losses in the distance, d_i. Where the current mix along the path includes sections of both warm sea and cold sea zones, all the sea along that path is assumed to be warm sea.

For the main or supplementary contour:

$$L_1(p) = L_b(p) - A_h \tag{27a}$$

For an auxiliary contour:

$$L_{1q}(p) = L_{bq}(p) - A_h \tag{27b}$$

where:

$L_b(p)$ (dB) and $L_{bq}(p)$ (dB): minimum required loss required for p% of the time for the main or supplementary contour and the auxiliary contour with value Q (dB), respectively (see equation (22)).

Iterative calculations

At the start of each iteration calculate the current distance for $i = 0, 1, 2,\ldots$:

$$d_i = d_{min} + i \cdot s \tag{28a}$$

The correction factor, C_i (dB), (see § 4.4 of the main body of this Appendix) for the distance, d_i, is given by:

$$C_i = \begin{cases} Z(f)(d_i - d_{min}) & \text{dB} \quad \text{for the main or supplementary contour} \\ 0 & \text{dB} \quad \text{for the auxiliary contour} \end{cases} \tag{28b}$$

where $Z(f)$ is given by equation (21) in § 4.4 of the main body of this Appendix.

At distances greater than 375 km, the value of the correction factor (C_i in equation (28b)) to be applied is the value of C_i at the 375 km distance.

The loss, $L_{bl}(p)$, where it is assumed that the path is wholly land (Zones A1 or A2), is evaluated successively using:

$$L_{bl}(p) = 142.8 + 20\log f + 10\log p + 0.1\,d_i + C_i \tag{29}$$

The loss, $L_{bs}(p)$, where it is assumed that the path is wholly cold sea (Zone B) or warm sea (Zone C), is evaluated successively using:

$$L_{bs}(p_1) = \begin{cases} 49.91\,\log\left(d_i + 1840\,f^{1.76}\right) + 1.195 f^{0.393}(\log p)^{1.38} d_i^{0.597} \\ +\,(0.01\,d_i - 70)\,(f - 0.1581) + (0.02 - 2\times10^{-5}\,p^2)\,d_i \\ +\,9.72\times10^{-9}\,d_i^{\,2}\,p^2 + 20.2 \end{cases} \text{ for Zone B} \tag{30a}$$

$$\begin{cases} 49.343\,\log\left(d_i + 1840\,f^{1.58}\right) + 1.266\,(\log p)^{(0.468+2.598\,f)}\,d_i^{0.453} \\ +\,(0.037 d_i - 70)\,(f - 0.1581) + 1.95\times10^{-10} d_i^{\,2} p^3 + 20.2 \end{cases} \text{ for Zone C} \tag{30b}$$

The predicted path loss at the distance considered is then given by:

$$L_2(p) = L_{bs}(p) + \left[1 - \exp\left(-5.5\left(\frac{d_{tm}}{d_i}\right)^{1.1}\right)\right]\left(L_{bl}(p) - L_{bs}(p)\right) \tag{31}$$

where:

d_{tm} (km): longest continuous land (inland + coastal) distance, i.e. Zone A1 + Zone A2 along the current path.

3 Frequencies between 790 MHz and 60 GHz

The propagation model given in this section is limited to an average annual time percentage (p) in the range 0.001% to 50%.

An iterative process is used to determine the propagation mode (1) required distance. First, equations (33) to (42) are evaluated. Then, commencing at the minimum coordination distance, d_{min}, equations (43) to (53) are iterated for distances d_i, where i = 0, 1, 2, ..., incremented in steps of s (km) as described in § 1.3 of the main body of this Appendix. For each iteration, d_i is the distance considered. This process is continued until either of the following expressions becomes true:

$$\begin{array}{llll} (L_5(p) \geq L_3(p)) & \text{and} & (L_6(p) \geq L_4(p)) & \text{for the main or supplementary contour} \\ (L_5(p) \geq L_{3q}(p)) & \text{and} & (L_6(p) \geq L_{4q}(p)) & \text{for the auxiliary contour} \end{array} \tag{32a}$$

or:

$$d_i \geq \begin{cases} d_{max1} & \text{for the main or supplementary contour} \\ d_1 & \text{for the auxiliary contour} \end{cases} \tag{32b}$$

The required distance, d_1, or the auxiliary contour distance, d_q, is then given by the current distance for the last iteration, i.e.:

$$d_1 = d_i \tag{32c}$$

or:

$$d_q = d_i \tag{32d}$$

Specific attenuation due to gaseous absorption

Calculate the specific attenuation (dB/km) due to dry air:

$$\gamma_o = \begin{cases} \left[7.19 \times 10^{-3} + \dfrac{6.09}{f^2 + 0.227} + \dfrac{4.81}{(f-57)^2 + 1.50} \right] f^2 \times 10^{-3} & \text{for } f \leq 56.77 \text{ GHz} \quad (33a) \\ 10 & \text{for } f > 56.77 \text{ GHz} \quad (33b) \end{cases}$$

The specific attenuation due to water vapour is given as a function of ρ (the water vapour density (g/m^3)) by the following equation:

$$\gamma_w(\rho) = \left(0.050 + 0.0021\rho + \frac{3.6}{(f-22.2)^2 + 8.5} \right) f^2 \rho \times 10^{-4} \tag{34}$$

Calculate the specific attenuation (dB/km) due to water vapour for the troposcatter propagation model using a water vapour density of 3.0 g/m^3:

$$\gamma_{wt} = \gamma_w(3.0) \tag{35a}$$

Calculate the specific attenuation (dB/km) due to water vapour for the ducting propagation model using a water vapour density of 7.5 g/m^3 for paths over land, Zones A1 and A2, using:

$$\gamma_{wdl} = \gamma_w(7.5) \tag{35b}$$

Calculate the specific attenuation (dB/km) due to water vapour for the ducting propagation model using a water vapour density of 10.0 g/m^3 for paths over sea, Zones B and C, using:

$$\gamma_{wds} = \gamma_w\,(10.0) \tag{35c}$$

Note that the value of 10 g/m^3 is used for both Zones B and C in view of the lack of data on the variability of water vapour density on a global basis, particularly the minimum values.

Calculate the frequency-dependent ducting specific attenuation (dB/km):

$$\gamma_d = 0.05 f^{1/3} \tag{36}$$

For the ducting model

Calculate the reduction in attenuation arising from direct coupling into over-sea ducts (dB):

$$A_c = \frac{-6}{(1 + d_c)} \tag{37}$$

where d_c (km) is the distance from a land based earth station to the coast in the direction being considered.

d_c is zero in other circumstances.

Calculate the minimum loss to be achieved within the iterative calculations:

$$A_1 = 122.43 + 16.5 \log f + A_h + A_c \tag{38}$$

For the main or supplementary contour:

$$L_3(p) = L_b(p) - A_1 \tag{39a}$$

For an auxiliary contour:

$$L_{3q}(p) = L_{bq}(p) - A_1 \tag{39b}$$

where:

$L_b(p)$ (dB) and $L_{bq}(p)$ (dB): minimum required loss required for p% of the time for the main or supplementary contour and the auxiliary contour with value Q (dB) respectively (see equation (22)).

For the tropospheric scatter model

Calculate the frequency-dependent part of the losses (dB):

$$L_f = 25 \log(f) - 2.5 \left[\log\left(\frac{f}{2} \right) \right]^2 \tag{40}$$

Calculate the non-distance-dependent part of the losses (dB):

$$A_2 = 187.36 + 10\varepsilon_h + L_f - 0.15\,N_0 - 10.1\left(-\log\left(\frac{p}{50}\right)\right)^{0.7} \quad (41)$$

where:

ε_h: earth station horizon elevation angle (degrees)

N_0: path centre sea level surface refractivity (see equation (11), § 4.1 in the main body of this Appendix).

Calculate the minimum required value for the distance dependent losses (dB):

For the main, or supplementary, contour:

$$L_4(p) = L_b(p) - A_2 \quad (42a)$$

For an auxiliary contour:

$$L_{4q}(p) = L_{bq}(p) - A_2 \quad (42b)$$

where:

$L_b(p)$ (dB) and $L_{bq}(p)$ (dB): minimum required loss required for p% of the time for the main or supplementary contour and the auxiliary contour of value Q (dB) respectively (see equation (22)).

Iterative calculations

At the start of each iteration, calculate the distance considered for $i = 0, 1, 2, ..$:

$$d_i = d_{min} + i \cdot s \quad (43)$$

Calculate the specific attenuation due to gaseous absorption (dB/km):

$$\gamma_g = \gamma_o + \gamma_{wdl}\left(\frac{d_t}{d_i}\right) + \gamma_{wds}\left(1 - \frac{d_t}{d_i}\right) \quad (44)$$

where:

d_t (km): current aggregate land distance, Zone A1 + Zone A2, along the current path.

Calculate the following zone-dependent parameters:

$$\tau = 1 - \exp\left[-\left(4.12 \times 10^{-4}\,(d_{lm})^{2.41}\right)\right] \quad (45)$$

where:

d_{lm} (km): longest continuous inland distance, Zone A2, along the path considered;

$$\mu_1 = \left[10^{\frac{-d_{tm}}{16\,-\,6.6\,\tau}} + \left[10^{-(0.496\,+\,0.354\,\tau)}\right]^5\right]^{0.2} \quad (46)$$

where:

d_{tm} (km): longest continuous land (i.e. inland + coastal) distance, Zone A1 + Zone A2 along the path considered.

μ_1 shall be limited to $\mu_1 \leq 1$.

$$\sigma = -0.6 - 8.5 \times 10^{-9} \; d_i^{3.1} \; \tau \tag{47}$$

σ shall be limited to $\sigma \geq -3.4$.

$$\mu_2 = \left(2.48 \times 10^{-4} \; d_i^2\right)^{\sigma} \tag{48}$$

μ_2 shall be limited to $\mu_2 \leq 1$.

$$\mu_4 = \begin{cases} 10^{(-0.935 + 0.0176 \; \zeta_r) \; \log \mu_1} & \text{for} \quad \zeta_r \leq 70° \quad (49a) \\ 10^{0.3 \log \mu_1} & \text{for} \quad \zeta_r \leq 70° \quad (49b) \end{cases}$$

where ζ_r is given in equations (9) and (10), § 4.1 in the main body of this Appendix.

Calculate the path-dependent incidence of ducting, β, and a related parameter, Γ_1 used to calculate the time dependency of the path loss:

$$\beta = \beta_e \cdot \mu_1 \cdot \mu_2 \cdot \mu_4 \tag{50}$$

where β_e is given in equations (7) and (8), § 4.1 in the main body of this Appendix.

$$\Gamma_1 = \frac{1.076}{(2.0058 - \log \beta)^{1.012}} \; \exp \left[-\left(9.51 - 4.8 \log \beta + 0.198 \, (\log \beta)^2\right) \times 10^{-6} \, d_i^{1.13} \right] \tag{51}$$

Calculate the correction factor, C_{2i} (dB) (see § 4.4 in the main body of this Appendix) using:

$$C_{2i} = \begin{cases} Z(f)(d_i - d_{min})\tau & \text{dB} \quad \text{for the main or supplementary contour} \\ 0 & \text{dB} \quad \text{for the auxiliary contour} \end{cases} \tag{52}$$

where $Z(f)$ is calculated using equation (21) in § 4.4 in the main body of this Appendix.

At distances greater than 375 km the value of the correction factor C_{2i} in equation (52) to be applied is the value of C_{2i} at the 375 km distance.

Calculate the distance-dependent part of the losses (dB) for ducting:

$$L_5(p) = (\gamma_d + \gamma_g)\ d_i + (1.2 + 3.7 \times 10^{-3}\, d_i) \log\left(\frac{p}{\beta}\right) + 12\left(\frac{p}{\beta}\right)^{\Gamma_1} + C_{2i} \tag{53}$$

and for tropospheric scatter:

$$L_6(p) = 20 \log\,(d_i) + 5.73 \times 10^{-4}\,(112 - 15\cos(2\zeta))\, d_i + (\gamma_o + \gamma_{wt})\, d_i + C_{2i} \tag{54}$$

For the determination of distances for auxiliary contours, C_{2i} = 0 dB.

4 Frequencies between 60 GHz and 105 GHz

This propagation model is valid for average annual percentage time (p) in the range from 0.001% to 50%.

An iterative process is used to determine the propagation mode (1) required distance. First, equations (55) to (59) are evaluated. Then commencing at the minimum coordination distance, d_{min}, equations (60) and (61) are iterated for distances d_i, where i = 0, 1, 2,..., incremented in steps of s (km) as described in § 1.3 of the main body of this Appendix. For each iteration, d_i is the distance considered.

This process is continued until either of the following expressions becomes true:

$$L_9(p) \geq \begin{cases} L_8(p) & \text{for the main or supplementary contour} \\ L_{8q}(p) & \text{for the auxiliary contour} \end{cases} \tag{54a}$$

or:

$$d_i \geq \begin{cases} d_{max1} & \text{for the main or supplementary contour} \\ d_1 & \text{for the auxiliary contour} \end{cases} \tag{54b}$$

The required distance, d_1, or the auxiliary contour distance d_q are then given by the current distance for the last iteration: i.e.

$$d_1 = d_i \tag{54c}$$

or:

$$d_q = d_i \tag{54d}$$

Calculate the specific attenuation (dB/km) for dry air in the frequency range 60 GHz to 105 GHz using:

$$\gamma_{om} = \begin{cases} \left[2 \times 10^{-4} \left(1 - 1.2 \times 10^{-5}\, f^{1.5}\right) + \dfrac{4}{(f - 63)^2 + 0.936} + \right. \\ \left. + \dfrac{0.28}{(f - 118.75)^2 + 1.771} \right] f^2\, 6.24 \times 10^{-4} & \text{dB/km} \quad \text{for} \quad f > 63.26 \text{ GHz} \quad (55a) \\ 10 \quad \text{dB/km} & \text{for} \quad f \leq 63.26 \text{ GHz} \quad (55b) \end{cases}$$

Calculate the specific attenuation (dB/km) for an atmospheric water vapour density of 3 g/m^3 using:

$$\gamma_{wm} = (0.039 + 7.7 \times 10^{-4} f^{0.5}) f^2 \; 2.369 \times 10^{-4} \tag{56}$$

Calculate a conservative estimate of the specific attenuation (dB/km) for gaseous absorption using:

$$\gamma_{gm} = \gamma_{om} + \gamma_{wm} \quad \text{dB/km} \tag{57}$$

For the required frequency and the value of earth station site shielding, A_h (dB), as calculated using the method described in § 1 of this Annex, calculate the minimum loss to be achieved in the iterative calculations:

$$L_7(p) = 92.5 + 20 \log (f) + A_h \quad \text{dB} \tag{58}$$

For the main or supplementary contour:

$$L_8(p) = L_b(p) - L_7 \quad \text{dB} \tag{59a}$$

For an auxiliary contour:

$$L_{8q}(p) = L_{bq}(p) - L_7 \quad \text{dB} \tag{59b}$$

where:

$L_b(p)$ (dB) and $L_{bq}(p)$ (dB): minimum required loss required for p% of the time for the main or supplementary contour and the auxiliary contour of value Q (dB) respectively (see equation (22)).

Iterative calculations

At the start of each iteration calculate the distance for $i = 0, 1, 2, ...$:

$$d_i = d_{min} + i \cdot s \tag{60}$$

Calculate the distance-dependent losses for the distance:

$$L_9(p) = \gamma_{gm} d_i + 20 \log (d_i) + 2.6 \left[1 - \exp \left(\frac{-d_i}{10} \right) \right] \log \left(\frac{p}{50} \right) \tag{61}$$

For frequencies above 60 GHz, the correction factor (see § 4.4 in the main body of this Appendix) is 0 dB. Therefore, no correction term has been added to equation (61).